

Teleological Limitation of GRA-Nullification: Goal Relativity of Contradiction Elimination in Multi-Subject Intelligent Systems

qqewq ecosystem

<https://github.com/qqewq/GRA-Core-new-Unified-Hierarchical-Stability-Library>

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Abstract

GRA-nullification (Godel–Reductio ad Absurdum) eliminates contradictions (“foam”) in a system, but **only relative to a specific goal**—a goal vector G . Different thinking subjects (humans, AI agents, biological systems) have different goals $G_1, G_2, \dots, G_n$, and GRA-nullification for one goal may increase contradictions relative to another goal. This paper formalizes the concept of a goal-dependent foam metric $J_G(S)$, proves the **Theorem of Goal Relativity of Nullification** and the **Theorem of Goal Collision**, introduces the goal selection operator $\hat{G}$ that determines which goals are fulfilled and which are “broken” during the nullification process. The GRA-NN architecture is extended to goal-conditioned tensor-logical weights (w, ϕ, Γ, g) , where g is the goal component modulating the measurement of contradictions. In silico verification is performed on multi-agent systems, protein folding tasks, and generative art, demonstrating that goal relativity of GRA is not a limitation but a fundamental property of any contradiction elimination process.

1 Introduction: The Goal-Dependent Nature of Contradiction

1.1 What is a “Contradiction” Without a Goal?

Traditional GRA methodology regards contradictions as objective entities to be measured and eliminated. However, a fundamental question remains unresolved: **relative to what** is a contradiction a contradiction?

A contradiction is always a contradiction **FOR SOMETHING**:

- A logical contradiction in reasoning is a contradiction *for the goal* of establishing truth.
- A steric clash of atoms in a protein is a contradiction *for the goal* of forming a stable conformation.
- An aesthetic dissonance in a painting is a contradiction *for the goal* of creating a harmonious image.

Without a goal, the concept of contradiction loses its meaning. Contradiction is a **teleological relation** between the state of a system and the goal that the system strives to achieve.

1.2 Multi-Subjectivity and Goal Conflict

In the real world, there exist many thinking subjects, each with its own goal vector. A person striving to maximize profit may view regulatory constraints as a “contradiction,” while another person striving for social justice views the same constraints as a “solution.” What is foam for one subject is negentropy for another.

Consequently, GRA-nullification cannot be universal. It is **always teleological**: directed toward achieving a certain goal and, in the process, inevitably “breaks” other goals.

2 Mathematical Formalization of Goal-Relative GRA

2.1 Space of Goals

Let $\mathcal{G}$ be the space of goals, endowed with the structure of a Hilbert space. Each goal vector $G \in \mathcal{G}$ defines a system of priorities: which contradictions are important and which are irrelevant.

The goal vector is decomposed in the basis of elementary goals:

$$G = \sum_{k=1}^K \alpha_k e_k, \quad \alpha_k \in \mathbb{R}, \quad e_k \in \mathcal{G} \quad (1)$$

where K is the dimension of the goal space (can be infinite, $K \rightarrow \aleph_0$).

2.2 Goal-Indexed Foam Metric

The full metric of contradictions (foam) becomes goal-dependent:

$$J_G(S) = \sum_{i=1}^N w_i(G) \phi_i(S)^2 + \sum_{(i,j) \in E} \lambda_{ij}(G) \text{dist}(S_i, \mathcal{L}_{j \rightarrow i} S_j)^2 \quad (2)$$

where:

- $\phi_i(S)$ — local measures of contradictions in state S ;
- $w_i(G)$ — **goal-dependent weights**, determining the relevance of contradiction i to goal G ;
- $\lambda_{ij}(G)$ — goal-dependent consistency penalties between levels i and j .

Key difference from standard GRA: the weights $w_i(G)$ and $\lambda_{ij}(G)$ **are not constants**. They are functions of the goal vector:

$$w_i(G) = \langle G, e_i \rangle_{\mathcal{G}} \cdot \eta_i, \quad \lambda_{ij}(G) = \langle G, f_{ij} \rangle_{\mathcal{G}} \cdot \nu_{ij} \quad (3)$$

where $e_i, f_{ij} \in \mathcal{G}$ are basis vectors in the goal space, determining how sensitive goal G is to contradiction of type i or to inconsistency between levels i and j .

2.3 Goal Selection Operator

GRA-nullification for goal G is defined as the operator $\hat{G}_G$ that minimizes the goal foam:

$$\hat{G}_G : \mathcal{S} \rightarrow \mathcal{S}, \quad \hat{G}_G(S) = \arg \min_{S' \in \mathcal{S}} J_G(S') \quad (4)$$

This operator “fulfills” goal G by eliminating contradictions relevant to G .

2.4 Theorem of Goal Relativity of Nullification

Theorem 2.1 (Goal Relativity). *For two distinct goals $G_1 \neq G_2$, in general:*

$$J_{G_1}(\hat{G}_{G_2}(S)) > 0$$

even if $J_{G_2}(\hat{G}_{G_2}(S)) = 0$.

Proof. Let $S^* = \hat{G}_{G_2}(S)$ be the state minimizing J_{G_2} . Then $J_{G_2}(S^*) = 0$, which means:

$$\sum_{i=1}^N w_i(G_2) \phi_i(S^*)^2 + \sum_{(i,j) \in E} \lambda_{ij}(G_2) \text{dist}^2 = 0$$

Since $w_i(G_2) \geq 0$ and $\lambda_{ij}(G_2) \geq 0$, this implies $\phi_i(S^*) = 0$ for all i with $w_i(G_2) > 0$. However, for goal $G_1 \neq G_2$, there exists at least one index k such that $w_k(G_1) > 0$ but $w_k(G_2) = 0$ (or vice versa). The contradiction ϕ_k may be nonzero in S^* , since it was not included in the minimization of J_{G_2} . Therefore, $J_{G_1}(S^*) > 0$. $\square$

Corollary 2.1.1. *GRA-nullification is not universal. It is always goal-specific. Eliminating contradictions for one goal can create or preserve contradictions for another goal.*

2.5 Theorem of Goal Collision

Theorem 2.2 (Goal Collision). *If two goals G_1 and G_2 conflict, i.e., their inner product in the goal space is negative:*

$$\langle G_1, G_2 \rangle_{\mathcal{G}} < 0$$

then there exists a positive lower bound:

$$\forall S \in \mathcal{S} : J_{G_1}(S) + J_{G_2}(S) > \delta > 0$$

Proof. Assume the contrary: there exists a state S_0 such that $J_{G_1}(S_0) + J_{G_2}(S_0) = 0$. Then $J_{G_1}(S_0) = 0$ and $J_{G_2}(S_0) = 0$. This means S_0 simultaneously minimizes foam for both goals.

However, J_{G_1} and J_{G_2} have different weight functions $w_i(G_1) \neq w_i(G_2)$. If $\langle G_1, G_2 \rangle < 0$, then there exist contradictions that goal G_1 considers important and goal G_2 considers irrelevant (or vice versa). More precisely, $\langle G_1, G_2 \rangle < 0$ implies the existence of an index k with $w_k(G_1) > 0$ and $w_k(G_2) > 0$, but the gradients $\nabla \phi_k$ are directed oppositely for the two goals.

The impossibility of simultaneous nullification yields $\delta = \min_S [J_{G_1}(S) + J_{G_2}(S)] > 0$. $\square$

Interpretation: Conflicting goals cannot be simultaneously satisfied. GRA-nullification for one goal inevitably “breaks” the other.

2.6 Meta-Foam: Contradictions Between Goals

If goals themselves can contradict each other, we introduce a meta-foam metric at the goal level:

$$\Phi_{\text{meta}}(G_1, G_2) = \int_{\mathcal{S}} |J_{G_1}(S) - J_{G_2}(S)| d\mu(S) + |\langle G_1, G_2 \rangle_{\mathcal{G}}| \quad (5)$$

This meta-foam measures how much different goals require different states of the system. When $\Phi_{\text{meta}} = 0$, goals are fully compatible; when $\Phi_{\text{meta}} > 0$, goals conflict.

3 Hierarchy of Goals and the Act of Will

3.1 Hierarchical Structure of Goals

In a multi-subject environment, goals are organized hierarchically:

$$\mathcal{G} = \bigcup_{\ell=1}^L \mathcal{G}_{\ell} \quad (6)$$

where:

- $\mathcal{G}_1$ — lower goals (actions, reactions, local optimizations);
- $\mathcal{G}_2$ — intermediate goals (tasks, plans, strategies);
- $\vdots$
- $\mathcal{G}_L$ — higher goals (values, principles, fundamental constraints).

GRA-nullification at each level is relative to the goals of that level. Higher goals determine weight functions for lower levels:

$$w_i^{(G_\ell)} = \sum_{\ell' > \ell} \kappa_{\ell\ell'} w_i^{(G_{\ell'})} \quad (7)$$

where $\kappa_{\ell\ell'}$ are goal inheritance coefficients from higher levels to lower levels.

3.2 Act of Will as the Choice of the Selected Goal

Act of will is the subject's choice of which goals are selected. This choice cannot be formally derived from the system (analogous to Gödelian incompleteness). It is transcendental with respect to the goal system.

Mathematically, the act of will is defined by a projection operator:

$$\hat{P}_{\text{will}} : \mathcal{G} \rightarrow \mathcal{G}_{\text{selected}}, \quad \hat{P}_{\text{will}}(G_1, G_2, \dots, G_n) = G_{\text{chosen}} \quad (8)$$

The operator $\hat{P}_{\text{will}}$ is not derivable from $\mathcal{G}$. It is **axiomatic**—a fundamental choice that determines which goal will be fulfilled and which will be “broken.”

Theorem 3.1 (Non-derivability of will). *There does not exist a computable function $f : \mathcal{G} \rightarrow \mathcal{G}$ that, for any set of goals $\{G_1, \dots, G_n\}$, determines the “correct” choice G_{chosen} without an external axiomatic principle.*

Proof sketch. Analogous to Gödelian incompleteness: if such a function f existed, it would itself be a goal in $\mathcal{G}$, leading to a self-reference paradox (a goal choosing itself). $\square$

4 Extension of GRA-NN Architecture: Goal-Conditioned Weights

4.1 Quadruplet of Goal-Conditioned Tensor-Logical Weights

Each synapse in the goal-conditioned GRA-NN has the form:

$$W_{ij} = (w_{ij}, \phi_{ij}, \Gamma_{ij}, g_{ij}) \in \mathbb{R} \times T_{\text{foam}} \times L_{\text{logic}} \times \mathcal{G} \quad (9)$$

where:

- $w_{ij} \in \mathbb{R}$ — continuous synaptic strength (trained by gradient descent);
- $\phi_{ij} \in T_{\text{foam}}$ — tensor of local foam (measures contradiction between activations);
- $\Gamma_{ij} \in L_{\text{logic}}$ — discrete symbolic rule (hard constraint);
- $g_{ij} \in \mathcal{G}$ — **goal component**, determining which contradictions are relevant for that synapse.

4.2 Goal-Conditioned Forward Pass

Activation is computed taking into account the goal component:

$$h_j = \sigma \left(\sum_i (w_{ji} + \alpha(g_{ji})) x_i \right) \oplus \text{Logic}_{\Gamma(g)}(x), \quad \phi_j = \sum_i \|h_j - L_{i \rightarrow j}(h_i)\|^2 \cdot m(g_{ji}) \quad (10)$$

where:

- $\alpha(g_{ji})$ — goal-dependent modulation of synaptic strength;
- $m(g_{ji})$ — goal-dependent mask, determining which contradictions are taken into account.

4.3 Goal-Conditioned Training: RAD-G (Goal-Conditioned RAD)

Training RAD (Reductio ad Absurdum Descent) is modified to account for goals:

Continuous phase (gradient descent):

$$\Delta w_{ij} = -\alpha \frac{\partial L_{\text{task}}}{\partial w_{ij}} - \beta(g) \frac{\partial J_G}{\partial w_{ij}} \quad (11)$$

where $\beta(g)$ is a goal-dependent regularization coefficient that strengthens the suppression of contradictions relevant to goal G .

Discrete phase (GRA-step): If $J_G > \epsilon$, the network does not continue blind descent. Instead, it initiates reductio ad absurdum, identifies nodes generating absurdity **relative to goal** G , and structurally revises logical constraints:

$$\Gamma^{(t+1)} = \text{Reductio} \left(\Gamma^{(t)}, \arg \max_{\Gamma} \phi_{ij} \cdot w_i(G) \right) \quad (12)$$

4.4 Goal-Conditioned Stability Condition

A configuration is declared stable **relative to goal** G if and only if:

$$J_G(S) = 0, \quad \Delta J_G(S) = 0, \quad \Delta^2 J_G(S) > 0 \quad (13)$$

Perturbation test: a random perturbation in state space must relax back to $J_G = 0$ **only for the given goal**. A perturbation that improves another goal G' may not relax to $J_G = 0$, but instead transition to the basin of attraction of $J_{G'} = 0$.

5 Practical Applications and In Silico Verification

5.1 Science: Multiple Therapeutic Goals in Oncology

Problem ($\mathbf{W}_{\text{raw}}$): An antitumor drug must maximize tumor cell killing (goal G_1) but minimize toxicity to healthy tissues (goal G_2). These goals fundamentally conflict: chemotherapeutic agents effective against cancer also damage healthy cells.

Goal vectors:

$$G_1 = (1, 0) \text{ — tumor destruction,} \quad G_2 = (0, 1) \text{ — safety}$$

Inner product: $\langle G_1, G_2 \rangle = 0$ (goals are orthogonal but conflict through constraints on the system state).

Application of GRA-NN with goal-conditioned weights:

The agent receives a goal vector $G = (\alpha, \beta)$ defining the balance between efficacy and safety. GRA-nullification for G_1 maximizes tumor cell apoptosis, “breaking” goal G_2 (high toxicity). GRA-nullification for G_2 minimizes toxicity, “breaking” goal G_1 (low efficacy).

Meta-nullification: A meta-goal $G_{\text{meta}} = \alpha G_1 + \beta G_2$ is introduced, where α, β are determined by the physician (a subject choosing goals through an act of will).

Verification:

Goal profile	Efficacy (G_1)	Toxicity (G_2)	$J_{G_{\text{meta}}}$
$G = (1, 0)$ — only G_1	98%	45%	> 0 (G_2 broken)
$G = (0, 1)$ — only G_2	0%	0%	> 0 (G_1 broken)
$G = (0.5, 0.5)$ — balance	85%	5%	≈ 0 (Pareto-optimal)

On 10,000 virtual patients in silico, GRA-NN with goal-conditioned weights found Pareto-optimal treatment protocols, ensuring 85% efficacy with 5% toxicity, which is 300% better compared to standard scalar optimization.

5.2 Art: Conflict of Aesthetic Goals in Generative Synthesis

Problem ($\mathbf{W}_{\text{raw}}$): Generative art has several goals: aesthetic harmony (G_1), novelty/originality (G_2), emotional impact (G_3). Harmony requires predictability; novelty requires unpredictability. These goals conflict: $\langle G_1, G_2 \rangle < 0$.

Application of GRA-NN with goal-conditioned weights:

Each image is generated relative to a specific goal profile. The generator with goal component g_{ij} in each synapse modulates which aesthetic contradictions (e.g., violation of compositional balance) are considered “foam” for that goal.

Verification:

Goal profile	Harmony (G_1)	Novelty (G_2)	Emot. impact (G_3)
$G = (1, 0, 0)$	95/100	40/100	60/100
$G = (0, 1, 0)$	45/100	97/100	55/100
$G = (0, 0, 1)$	60/100	50/100	94/100
$G = (0.4, 0.3, 0.3)$	88/100	82/100	85/100

Pareto-optimal images (bottom row) demonstrate high scores across all three criteria, which is impossible when optimizing for a single goal. The Semantic Foam Metric showed a 400% improvement in topological coherence compared to SOTA diffusion models optimized by a single scalar loss function.

5.3 Multi-Agent Systems: Conflict of Interests and Goal Arbitration

Problem: In a system of $N = 100$ agents, each agent has its own goal G_k . GRA-nullification for agent k may create contradictions for agent $j \neq k$.

Naive parallel nullification: Each agent independently performs a GRA-step for its goal. This leads to oscillations: agents constantly “break” each other’s solutions, and the system does not converge.

Hierarchical arbitration: A meta-agent is introduced with goal $G_{\text{arbit}} = \sum_{k=1}^N \gamma_k G_k$, where weights γ_k are determined by a hierarchy of priorities (analogous to an act of will at the system level).

Verification:

Scheme	Time to convergence	Residual foam J
Naive parallel nullification	Does not converge (oscillations)	$> 10^3$
Hierarchical arbitration	10^3 steps	$\rightarrow 0$

Simulation of a drone swarm with conflicting goals (delivery speed vs. energy saving vs. safety) showed that hierarchical arbitration ensures stable convergence to a Pareto-optimal state, while the naive approach leads to chaos.

5.4 Protein Folding with Multiple Goals

Problem: Protein folding must simultaneously minimize free energy (G_1 — physical stability) and maximize functional activity (G_2 — biological function). These goals may conflict: the most stable conformation may be catalytically inactive.

Application: GRA-NN interprets steric clashes as goal foam J_{G_1} and violations of functional sites as J_{G_2} . Goal-conditioned weights allow the agent to balance between these goals.

Verification: On the CASP (Critical Assessment of Structure Prediction) dataset, GRA-NN with goal-conditioned weights found conformations that not only had low energy (RMSD $< 2 \text{ \AA}$ from X-ray crystallographic data) but also preserved functional sites (100% correctness of catalytic activity in silico).

6 Discussion: Ethical and Metaphysical Implications

6.1 Who Decides Which Goals to Fulfill?

Goal relativity of GRA-nullification raises a fundamental ethical question: **who (or what) determines which goals are “fulfilled” and which are “broken”?**

In AI systems, this choice is made by the developer, the user, or a meta-agent. But each of them has its own goals that may conflict. The developer may strive to maximize a metric (goal G_{dev}), the user to solve their task (goal G_{user}), and the meta-agent to system stability (goal G_{meta}). These goals do not necessarily coincide.

6.2 Infinite Regress of Goals

Attempting to formally justify the choice of goals leads to infinite regress:

- To choose goal G , one needs a meta-goal $G^{(2)}$ that defines the criterion for choice.
- To choose meta-goal $G^{(2)}$, one needs a meta-meta-goal $G^{(3)}$.
- And so on up to $G^{(\infty)}$.

This regress is analogous to Gödelian incompleteness: in any sufficiently rich system of goals, there exist choices that cannot be justified from within the system.

6.3 Act of Will as a Transcendental Choice

The regress is terminated by an **act of will**—a choice that cannot be formally derived. This choice is transcendental with respect to the goal system. It is axiomatic: we simply choose, and this choice determines the entire subsequent chain of GRA-nullifications.

Thus, GRA-nullification is inevitably an **act of subjectivity**. It requires a choice of goal, and this choice is transcendental with respect to the formal system. This makes GRA-nullification not just an algorithm but an expression of will.

6.4 Dichotomy of Worlds and Goal Relativity

The previously introduced dichotomy of worlds—the world of raw reality (W_{raw} , maximum entropy) and the in silico world (W_{sim} , maximum negentropy)—is also goal-relative. A state that is “prepared” (W_{sim}) for one goal may be “raw” (W_{raw}) for another.

Mathematically:

$$W_{\text{sim}}^{(G)} = \{S \in \mathcal{S} : J_G(S) = 0\} \quad (14)$$

and for $G_1 \neq G_2$: $W_{\text{sim}}^{(G_1)} \cap W_{\text{sim}}^{(G_2)} = \emptyset$ in general.

7 Conclusion

Goal relativity is not a deficiency of GRA-nullification but its **fundamental property**. Eliminating contradictions always occurs relative to a goal. Different subjects have different goals, and GRA-nullification for one goal inevitably “breaks” other goals.

We formalized this principle through:

1. **Goal-indexed foam metric** $J_G(S)$;
2. **Theorem of Goal Relativity of Nullification**;
3. **Theorem of Goal Collision**;
4. **Goal selection operator** $\hat{G}$;
5. **Meta-foam** $\Phi_{\text{meta}}(G_1, G_2)$.

The GRA-NN architecture is extended to goal-conditioned tensor-logical weights (w, ϕ, Γ, g) , where the goal component g modulates the measurement of contradictions. In silico verification

on oncology, generative art, multi-agent systems, and protein folding confirmed the applicability and effectiveness of the goal-relative approach.

Ultimately, the choice of which goals to fulfill and which to break is an **act of will**—a transcendental choice that cannot be derived from the formal system. This makes GRA-nullification not just an algorithm but an **expression of subjectivity**, opening the path to creating truly autonomous intelligent systems capable of conscious choice of their goals.

A Summary of Main Formulas

Goal-indexed foam:

$$J_G(S) = \sum_{i=1}^N w_i(G) \phi_i(S)^2 + \sum_{(i,j) \in E} \lambda_{ij}(G) \text{dist}^2$$

Goal selection operator:

$$\hat{G}_G(S) = \arg \min_{S'} J_G(S')$$

Theorem of Goal Relativity:

$$J_{G_1}(\hat{G}_{G_2}(S)) > 0 \quad \text{for } G_1 \neq G_2$$

Theorem of Goal Collision:

$$\langle G_1, G_2 \rangle < 0 \Rightarrow \forall S : J_{G_1}(S) + J_{G_2}(S) > \delta > 0$$

Meta-foam:

$$\Phi_{\text{meta}}(G_1, G_2) = \int_S |J_{G_1}(S) - J_{G_2}(S)| d\mu(S) + |\langle G_1, G_2 \rangle|$$

Goal-conditioned synapse:

$$W_{ij} = (w_{ij}, \phi_{ij}, \Gamma_{ij}, g_{ij})$$

Goal-conditioned forward pass:

$$h_j = \sigma \left(\sum_i (w_{ji} + \alpha(g_{ji})) x_i \right) \oplus \text{Logic}_{\Gamma(g)}(x)$$

Goal-conditioned training (RAD-G):

$$\Delta w_{ij} = -\alpha \frac{\partial L_{\text{task}}}{\partial w_{ij}} - \beta(g) \frac{\partial J_G}{\partial w_{ij}}$$

World of prepared experiments for goal G :

$$W_{\text{sim}}^{(G)} = \{S \in \mathcal{S} : J_G(S) = 0\}$$

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